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Task 1: Research Boeing Drama

The recent Boeing Drama started in October with a Boeing 737 Max 8 plane flying for Lion Air crashing and killing everyone on board. The crash was attributed to pilot error. 5 months later on March 10th there was another 737 Max 8 crash, this one flying for Ethiopian Airlines.

After the second crash, Boeing announced the possibility of them being attributed to a new software system: the Maneuvering Characteristics Augmentation System (MCAS). The MCAS was designed to solve a problem that stemmed from the top; in the spirit of competition with the rival company Airbus, Boeing felt the need to release a plane with larger and more fuel efficient engines. This change posed a difficult choice: since the current 737 model had a shorter stature than the rival Airbus A320neo, Boeing needed to either design a whole new plane, or make substantial changes to the existing 737 model to incorporate a larger under-wing engine.

Due to cost and time limitations, Boeing decided to modify the existing design. Any changes to the geometry of an airplane has a cascading effect, calling for changes to many other aspects of the airplane. These changes affected the aerodynamics of the redesigned 737 Max 8 airplane, the altered center of gravity would cause the nose to tip upwards, which would increase the likelihood of a stalling situation. To counter-act this phenomenon, Boeing engineers designed the MCAS system, that would use multiple sensor readings to detect a potential deviation in the pitch angle and adjust the tail wing flaps and the jack screw to push the nose downwards again. The MCAS system primarily worked, but pilots reported that it would sometimes cause inconsistent handling situations. For such a drastic change in Pilot protocol, Boeing should have sponsored a wide-spread re-training program for any pilots flying with the new system. The prohibitive cost of new flight simulators and full training programs caused Boeing to instead issue short training videos and updated flight manuals.

On both of the doomed flights, there was evidence to suggest that the crashes were directly due to a nose-dive situation, which implicates the MCAS system, which was designed to do push the nose down in a bad situation. Obviously that system malfunctioned, and continued to push the nose down when it shouldn't have, which caused the pilots to lose control and the planes to crash straight into the ground. The malfunctions were due to a composition of faulty angle of attack sensors, and pilot error. Adding further problems, there was an optional package that included an extra angle of attack vane, and a warning light on the dash that would trigger if the two angle of attack vanes disagreed, warning the pilots that there may be a problem and giving them time to adjust manually. Neither of the crashed planes had this optional package installed.

From a general view, it could be seen that the software system was directly responsible for the crashes. Once we look into the details, we find that the software system (the MCAS) was designed to fix the problems in the electrical system (un-reliable angle of attack sensors), which was designed to fix a

mechanical system issue (inconsistent angle-of-attack adjustments), which was all designed to fix an aerodynamic issue (improper center-of-gravity and airplane geometry). All of these changes were inspired solely by economic motivations, which are triggered by corporate management at the top of the chain. The initial blame was placed on the Software system and the software developers that designed that system, and only under careful scrutiny has the rest of the tale unfolded.

On top of all of the systems problem, the decided factor for these crashes lied with the people piloting the plane. While it was not necessarily their fault, they had improper training and were kept ignorant of the implications of these systems. In previous flights, some pilots experienced similar problems with the nose-dive situation, and had the training or knowledge to deactivate the system and regain control, maybe with help from the optional warning light system. In the two crashes, the pilot's either did not have time, or did not have the necessary knowledge to deactivate the faulting system, which led to both fatal crashes.

One of the largest flaws in the Software is not a bug; it is the fact that the MCAS uses readings from only one angle of attack sensor, without a redundancy built in. The proposed fix from Boeing is changing the software to use two angle of attack sensors, with the previously-optional warning light a mandatory feature. This would allow for redundancy in the sensor system, if one of the sensors fails, which is an expected cause of at least one of the crashes, then the other sensor can be the backup. With the now-mandatory warning light, the pilot will now be aware if there is a partial sensor failure, and since they still have function from the other sensor, will have adequate time to respond and to ground the plane safely. Even though the failings started all the way up the line, this was the critical failing point of the software design. While the software did do what was intended, it could not function properly without proper input from the mechanical/electrical systems. And if the software is not functioning properly, and the pilot is unaware/untrained to handle that malfunction, then it was inevitable there will be a nuclear failure due to unforeseen circumstances like happened in the Ethiopian Airlines and Lion Air plane crashes.

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